

PetCuesGame – Educational Animal Safety Game

Project Check-in 2

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Project Demo

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Research Paper Progress

- Increase Engagement through Gamification – Use choice-based gameplay to make learning fun, interactive, and memorable.
- Ensure Accessibility and Scalability – Deliver the game through a web-based platform compatible with PCs, tablets, and mobile devices, allowing easy adoption in schools, humane societies, and home environments.
- Support United Nations Sustainable Development Goals (SDGs) – Align the project with SDG 3 (Good Health Well-being) by reducing animal-related injuries, SDG 4 (Quality Education) by providing an engaging educational tool, and SDG 15 (Life on Land) by promoting responsible pet care.

By leveraging technology and interactive learning, PetCues-Game aims to transform how animal safety education is delivered, making it accessible, engaging, and impactful for diverse communities. This paper explores the technological framework, educational benefits, and potential global impact of the project while addressing the challenges and future possibilities of improving animal safety education.

II. LITERATURE REVIEW

A. Game-Based Learning and Its Impact on Motivation

Game-based learning (GBL) is widely recognized for enhancing student motivation by providing interactive and immersive environments. Liao et al. (2009) argue that digital game-based learning environments foster both intrinsic motivation and engagement by allowing learners to interact with virtual companions while completing performance-based tasks. The authors introduce a Performance-Based Learning (PeBL) approach, which emphasizes two core dimensions—effort-making and identity-making. In the effort-making dimension, learners engage in back-stage tasks where they repeatedly practice skills through training their virtual pets. Meanwhile, the identity-making dimension involves front-stage performance, where learners showcase their progress, promoting self-regulation and boosting their motivation through peer evaluation (Liao et al., 2009, p. 430).

Similarly, She et al. (2013) demonstrate that social-network-based educational games on platforms like Facebook can sustain and improve learners' motivation. Their research on the "Operating System Little Pet (OSLP)" game indicates that incorporating social elements—such as virtual pet battles and leaderboards—encourages students to engage more actively with course material. The study found that students were willing to interact with the game outside regular class hours, and competitive elements significantly boosted their motivation (She et al., 2013, p. 226).

B. Virtual Pets as Educational Tools

The use of virtual pets as educational tools is an effective strategy to foster engagement and reinforce learning. According to Liao et al. (2009), the My-Pet and Our-Stage system allows learners to interact with virtual pets in a dual-mode learning environment. In the My-Pet system, learners nurture and train their pets through a series of problem-solving tasks,

while the Our-Stage system enables learners to present their pets in competitions and performances (Liao et al., 2009, p. 430). This dual approach integrates cognitive learning (through task completion) with social motivation (through public performance), creating a balanced environment that enhances both skill development and emotional engagement.

She et al. (2013) reinforce this concept by highlighting how the OSLP game uses virtual pets to connect course content with real-time decision-making. In their study, students demonstrated increased engagement and better performance when allowed to raise and compete with virtual pets. The social interaction created by pet battles and in-game leaderboards encouraged collaborative learning and motivated students to revisit and master the content (She et al., 2013, p. 226).

C. Social Interaction and Collaborative Learning

Both studies highlight the importance of social interaction in game-based learning. She et al. (2013) emphasize that social-network-based games create a collaborative learning environment where students can interact, compete, and cooperate asynchronously. This flexibility allows learners to engage with content at their own pace while remaining connected to their peers through competitive features like leaderboards and pet battles. The study showed that these social elements not only improved learning outcomes but also sustained long-term engagement (She et al., 2013, p. 226).

Liao et al. (2009) take this concept further by describing how performance-based learning provides opportunities for peer-to-peer feedback and public recognition. In the Our-Stage system, learners present their trained pets on a virtual stage, where audiences (other learners) observe, evaluate, and provide feedback. This continuous cycle of performance and evaluation fosters self-reflection and motivation. Moreover, the study found that the public nature of performances helps learners internalize knowledge through observation and participation in community-based learning (Liao et al., 2009, p. 430).

D. Relevance to the PetCuesGame Project

Insights from these studies are directly applicable to the design of the PetCuesGame:

- Virtual Pet System: Similar to the My-Pet and Our-Stage system, learners can nurture a pet by completing educational tasks related to animal safety.
- Social and Collaborative Learning: Implementing pet competitions and allowing children to interact with their peers can replicate the engagement benefits seen in both studies.
- Progressive Learning Pathways: The game should gradually increase in complexity, reinforcing learning through incremental challenges and immediate feedback.
- Identity and Effort Mechanisms: Incorporating public recognition through pet displays or leaderboards can mirror the identity-making process, fostering a sense of achievement and self-efficacy.

Research Paper Progress

III. GAME DESIGN AND EDUCATION FRAMEWORK

A. Core Elements of Game-Based Learning Design

a) Motivation and Engagement:

- Intrinsic Motivation: Games should foster intrinsic motivation by allowing players to nurture virtual pets, creating an emotional connection to learning (Liao et al., 2009, p. 430)
- Competitiveness and Social Interaction: Implementing competitive mechanics like pet battles and leaderboards encourages learners to stay engaged and revisit educational content (She et al., 2013, p. 226)
- Goal-Oriented Learning: Define clear goals and challenges that reflect real-life scenarios, such as safely approaching or handling animals.

b) Feedback and Assessment:

- Immediate Feedback: Both studies highlight the importance of providing real-time feedback. This helps learners correct mistakes and reinforces learning (She et al., 2013, p. 225).
- Formative Assessment: Use in-game quizzes or tasks that track a learner's progress while allowing them to practice repeatedly (Liao et al., 2009, p. 431).

B. Game Narrative and Contextual Learning

- Story-Driven Gameplay: Research suggests that embedding learning in a narrative context enhances knowledge retention. Liao et al. (2009) describe how performance-based learning through a virtual pet world makes educational tasks meaningful (p. 430).
- Real-Life Scenarios: She et al. (2013) show how embedding course content in interactive, relatable contexts helps learners apply knowledge outside the game environment (p. 226).

In PetCuesGame:

- Use a story-driven approach where children help lost pets or guide animals safely through the community.
- Incorporate scenarios like meeting a stray dog, where players must make safe choices (e.g., "Ask an adult for help").

IV. PREPARE YOUR PAPER BEFORE STYLING

Thank You